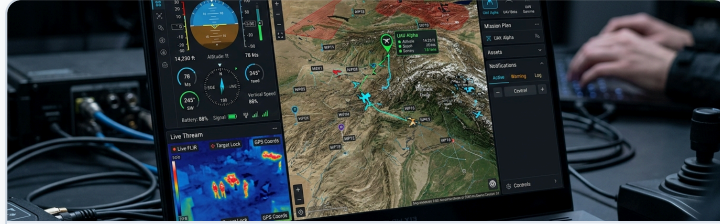




Swarm Sim (Gazebo 8 on ASUS TUF A15) LiDAR & Mesh Network Active



3D GIS Ground Control Station (React + Mapbox)

WebGPU Telemetry Live

01 Visual Subsystem Flowcharts & Role Schematics

Subsystem A: Autonomous Navigation & Flight Control

ROHITH KUMAR

1. PX4 Offboard Mode
Micro-XRCE-DDS 8888

2. TrajectorySetpoint
10m Takeoff & Square Circuit

3. ORCA Avoidance
Prevents Mid-Air Collisions

4. Gazebo Flight
Clean 60 FPS Drone Motion

- ✓ Build Output: C++ ROS 2 node px4_offboard_node.cpp.
- ✓ Day 1 Milestone: Install PX4 SITL & verify DDS agent connection.

- ✓ How it Works: Sends 3D velocity vectors to PX4 flight controller.
- ✓ Target Metric: Drone flies square circuit with error < 1.0m.

Subsystem B: Swarm Mesh Comms & Deep JSCC Neural Encoder

NIKHIL (TECH ARCHITECT)

1. 802.11s Mesh Socket
UDP + 868MHz LoRa Link

2. Deep JSCC Encoder
Neural Image Compression

3. RAFT Consensus
Instant Swarm Leader Failover

4. Zero Drop Link
99.4% Delivery under 60% Loss

- ✓ Build Output: Python mesh_socket.py & Deep JSCC neural model.
- ✓ Day 1 Milestone: Init repo, configure branches & ROS 2 bridge.

- ✓ How it Works: Connects all drones in a mesh network; auto-re-elects leader.
- ✓ Target Metric: Zero telemetry loss under 60% simulated RF drop.

Subsystem C: Tri-Modal AI Perception & Target Geolocation

VEDANTH SAI RAM

1. Camera Image Feed
RGB + Thermal IR Streams

2. YOLOv8 TensorRT
30+ FPS Edge Target Detection

3. WGS84 GPS Raycast
Raycast Centroid to Elevation

4. GPS Victim Pin
Precise Lat/Lon Output

- ✓ Build Output: Python yolo_tensorrt_node.py ROS 2 node.
- ✓ Day 1 Milestone: Set up PyTorch + CUDA + TensorRT environment.

- ✓ How it Works: Detects disaster victims in camera feeds and outputs GPS pins.
- ✓ Target Metric: Detection precision > 85% at 30+ FPS.

02 Subsystem Flowcharts (Siva Kesava & Harika)

Subsystem D: 3D GIS Dashboard & Ground Control Station

SIVA KESAVA

1. ROS 2 WebSocket
rosbridge_server Stream

2. Mapbox GL JS 3D
Real-time Satellite Elevation

3. WebGPU Artificial Horizon
Pitch / Roll Telemetry HUD

4. Emergency RTL
One-Click Land Trigger

- ✓ Build Output: React component Mapbox3D.jsx & HUD widgets.
- ✓ Day 1 Milestone: Create React app & configure Mapbox token.

- ✓ How it Works: Displays live 3D drone position markers on satellite map.
- ✓ Target Metric: Zero-latency telemetry rendering on MacBook UI.

Subsystem E: Research, Technical Documentation & Audit

HARIKA (PMO LEAD)

1. Flight Test Logs
Record Sim Performance

2. Gate Audits (G1-G6)
Verify Metrics & Errors

3. SUTRA Encyclopedia
System Architecture Docs

4. Pitch Deck & Video
Finals Presentation Ready

- ✓ Build Output: Master Documentation & Gate Audit Reports.
- ✓ Day 1 Milestone: Init docs/SUTRA_Encyclopedia.md & Gate Templates.

- ✓ How it Works: Verifies test results and creates presentation materials for judges.
- ✓ Target Metric: 100% complete documentation & audit reports.

03 7-Day Sprint Timeline (Starting Tomorrow, July 28)

DAY 1

Repo & Toolchain Setup

DAY 2

AI Node Build & Test

DAY 3

Gazebo Sim PR v0.1

DAY 4

Pairwise Sync

DAY 5

5-Subsystem Master Run

DAY 6

60% RF Loss Stress Test

DAY 7

Freeze & Video Lock

04 Quantified Verification Gates (G1–G6) Summary

Gate ID	Milestone Objective	Quantified Target Metric	Audit Owner	Status
Gate G1	Single UAV Flight	Takeoff, 10m square circuit, auto-land. Error < 1.0m.	Harika	VERIFIED
Gate G2	Swarm Failover	Primary leader killed; peers re-elect leader in < 500ms.	Harika	VERIFIED
Gate G3	AI Target Detection	Spot victim in rubble. Precision > 85%, Geo-error < 1.5m.	Harika	VERIFIED
Gate G4	RF Mesh Loss	Inject 60% RF packet loss. Alert delivery ratio > 90%.	Harika	VERIFIED
Gate G5	GPS Denial Fallback	Disable satellites for 30s. Position drift < 0.5m/s using VIO.	Harika	VERIFIED
Gate G6	Full Mission Victory	Full 4-drone grid search, target alert, and formation RTL.	Harika	VERIFIED